

Juice Spacecraft Uses Earth's Gravity to Head Toward Jupiter

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A **spacecraft** named Juice got a boost from Earth on September 28. Juice is short for Jupiter Icy Moons Explorer, a mission run by the European Space Agency, or ESA. It used Earth's gravity to bend its path toward Jupiter and gain speed. This is called a **gravity assist**, and it helps save fuel.

The flyby changed Juice's direction by 20 degrees and increased its speed by 3.5 kilometers per second. During the four weeks before the encounter, controllers needed only one small **course correction**. They had set aside six chances to adjust its path.

At its closest point, Juice passed 8,640 kilometers above the Indian Ocean. Its monitoring cameras took a series of pictures.

Angela Dietz, Juice's **Spacecraft** Operations Manager, explained that careful planning meant the team used only a small amount of the fuel reserved for this flyby. That leaves more fuel for studying Jupiter's icy moons later.

Changing direction was the main goal, but the flyby also let the team test Juice's science **instruments**. Earlier testing opportunities came during a lunar-Earth **gravity assist** in 2024 and observations of Comet 3I/ATLAS in 2025.

Teams spent months planning when each of Juice's 10 **instruments** could work. Project scientist Claire Vallat explained that the team had to choose which activities came first. For example, Juice relied only on battery power while in Earth's shadow. Tests that helped prepare the **instruments** for Jupiter received priority.

At Jupiter, Juice is scheduled to make 35 flybys of the large moons Ganymede, Callisto and Europa. Earth served as a **rehearsal** to help scientists prepare for those encounters.

Juice also spent several days traveling through Earth's **magnetotail**. This is the long part of Earth's magnetic field that stretches away from the Sun.

Word Treasure

spacecraft -- A vehicle designed to travel in space.

gravity assist -- Using a planet's gravity to change a spacecraft's path and speed.

course correction -- An adjustment to the path a spacecraft follows.

instruments -- Tools used to make scientific observations or measurements.

rehearsal -- Practice that helps prepare for a later activity.

magnetotail -- The long part of Earth's magnetic field stretching away from the Sun.

Background you need

A gravity assist uses a planet's gravity to change a spacecraft's path and speed.
Earth's magnetotail is part of its magnetic field and stretches away from the Sun.

Break it down

Introduces Juice and explains its gravity assist.
Reports changes in direction and speed, navigation work and the closest approach.
Explains fuel savings and science instrument testing.
Connects the Earth rehearsal to future moon flybys and describes the magnetotail.

Why it matters

The Earth flyby saved fuel and helped scientists test instruments before Juice studies Jupiter's icy moons.

Test what you remember

1. Which planet is Juice traveling toward?

- ☐ (A) Mars
- ☐ (B) Venus
- ☐ (C) Saturn
- ☐ (D) Jupiter

2. What helped Juice change its path and gain speed?

- ☐ (A) Earth's sunlight
- ☐ (B) Earth's oceans
- ☐ (C) Earth's gravity
- ☐ (D) Earth's clouds

3. How many small course corrections did controllers need in the four weeks before the flyby?

- ☐ (A) Two corrections
- ☐ (B) Four corrections
- ☐ (C) One correction
- ☐ (D) Six corrections

4. Which ocean was below Juice at its closest point to Earth?

- ☐ (A) The Pacific Ocean
- ☐ (B) The Indian Ocean
- ☐ (C) The Atlantic Ocean
- ☐ (D) The Arctic Ocean

5. What supplied Juice's power while it was in Earth's shadow?

- ☐ (A) Sunlight alone
- ☐ (B) Water power alone
- ☐ (C) Wind power alone
- ☐ (D) Battery power alone

6. How many flybys of Jupiter's large moons is Juice scheduled to make?

- ☐ (A) 20 flybys
- ☐ (B) 35 flybys
- ☐ (C) 50 flybys
- ☐ (D) 10 flybys

Answer key (try the quiz first!): 1: D 2: C 3: C 4: B 5: D 6: B

Questions to think about

1. **Angela Dietz:** She explained that careful planning reduced fuel use, leaving more for observations of Jupiter's icy moons.
2. **Claire Vallat:** She explained that the team prioritized instrument activities that helped prepare Juice for work at Jupiter.

MY THOUGHTS (at least 20 words):